JONATHAN NADEAU

DATA ANALYST

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Dear Hiring Manager,

I am a UCSB Mathematics graduate and Google Data Analytics certified analyst, currently completing UCLA Extension's Data Science Certificate. As the sole developer at Escape Communications, a defense and telecom hardware manufacturer in Torrance, I have designed, built, validated, and shipped four internal tools that engineers and executives now use in their work. That internship is the best evidence I can offer of how I operate as an analyst, so I want to walk you through it.

The first project began with a locked door. The company's failure-analysis history, 4,038 hardware failure records dating to 2007, sat in a legacy Bugzilla system whose administrator had departed years earlier without documentation. I reverse-engineered the undocumented data model, extracted the complete corpus with full comment history via REST API, and audited every record. That audit reshaped the project: the structured classification fields were more than 80% blank or defaulted, while the real engineering knowledge lived in free-text comment threads. Rather than build dashboards on empty fields, I built a deterministic Python pipeline that indexed 2,245 component reference designators out of unstructured text, hand-validated it prefix by prefix, and eliminated a 68 to 71% false-positive rate on the worst values before trusting a number. The finished Streamlit application gives engineers component-level failure history and similarity search the original system never offered, and it shipped with a runbook, user guide, and append-only build log so the knowledge cannot be lost the same way twice.

The second tool, the BOM Checker, now runs company-wide as a standalone Windows executable. It audits raw bill-of-materials exports and flags 16 classes of defect without modifying a single cell, a flag-don't-fix design that came from a working session with the VP of Engineering: his workflow is to correct errors at the source, so a tool that silently cleaned files would hide exactly what he needed to see. To catch the errors hardest for a human to spot, I built an offline decoder for more than ten manufacturers' part-numbering schemes and cross-checked what each part number actually encodes against what the BOM claims. Validated across eight production BOMs spanning five product lines, it surfaced 21 datasheet-confirmed spec errors, including a 100V requirement on a part rated for 16V, backed by 427 automated tests and byte-for-byte output verification.

The third, a BOM sourcing and purchase-planning pipeline, integrates live distributor APIs (Mouser production, DigiKey OAuth2) to produce cost-minimized, fully auditable multi-vendor buy plans. On a real 100-board build, volume pricing invisible to the prior manual process cut the parts plan from $129.8K to $103.8K, roughly 20%, and availability checks flagged 55 at-risk parts before the build rather than during it. I briefed the VP of Engineering in a one-page readout that led with the financial case.

The fourth I proposed myself. I asked an FPGA engineer where a data person could help, and identified that his waveform verification work was fundamentally a data comparison problem. I reverse-engineered the vendor capture archive, reconstructed the message stream from the hardware handshake signals, and shipped offline GUI and command-line executables that run 13 named checks and locate corruption to the exact word and sample index. When one new check failed against a capture known to be good, I investigated instead of loosening the check, which surfaced two previously undocumented hardware handshake behaviors.

I am early in my career, and this internship is my first industry analyst role; I will not pretend otherwise. What I bring alongside it is a portfolio of fifteen deployed projects across SQL, Python, Excel modeling, and Power BI, and 1,500+ hours of mathematics instruction that taught me to explain quantitative findings to any audience. I used AI coding tools as an accelerator throughout the internship; every architecture, data-source, and methodology decision was mine, and every claim above was verified against real data before anyone was asked to trust it.

I would welcome the chance to bring that same standard of rigor, honesty, and shipped results to your team. Thank you for your consideration.

Sincerely,
Jonathan Nadeau